

ASCII MASTER FLOW – PANEL 1/12: Mountain-hazard taxonomy

LANDSLIDE -> rock debris or earth movement

ROCKFALL -> detached rock falls / bounces / rolls

DEBRIS FLOW -> water-rich channelised mass

AVALANCHE / GLOF -> snowpack release / lake outburst flood

ASCII MASTER FLOW – PANEL 2/12: Susceptibility-hazard-risk ladder

SUSCEPTIBILITY -> WHERE terrain may fail

HAZARD -> probability timing intensity

EXPOSURE + VULNERABILITY + CAPACITY

RISK -> expected harm and disruption

ASCII MASTER FLOW – PANEL 3/12: Slope-failure mechanism

CONDITIONING -> geology slope weathering drainage land use

TRIGGER -> rain quake erosion cutting loading

FAILURE -> slide fall or flow

EXPOSURE -> route settlement service consequence

ASCII MASTER FLOW – PANEL 4/12: Cascade map

EARTHQUAKE / RAIN -> LANDSLIDE

LANDSLIDE -> RIVER BLOCKAGE

BLOCKAGE OR LAKE-DAM FAILURE -> OUTBURST FLOOD

CLOUD BURST -> FLASH FLOW + DEBRIS + SLOPE FAILURE

ASCII MASTER FLOW – PANEL 5/12: Institutional owner matrix

GSI -> landslide maps inventory research forecast development

DGRE / DRDO -> snow and avalanche warnings

NRSC / ISRO -> Earth observation and inventories

CWC / STATE / DISTRICT -> downstream water and public action

ASCII MASTER FLOW – PANEL 6/12: Glacial-lake monitoring rail

- 1 REPEAT SATELLITE SCREENING
- 2 LAKE DAM SLOPE ICE AND OUTFLOW CHARACTER
- 3 FIELD VALIDATION + DOWNSTREAM EXPOSURE
- 4 WARNING EVACUATION AND REVIEW

ASCII MASTER FLOW – PANEL 7/12: Zonation-to-decision bridge

HIGHER SUSCEPTIBILITY / HAZARD?

AVOID OR REQUIRE SITE INVESTIGATION

CONTROL LAND USE DRAINAGE CUTTING LOADING

MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK

ASCII MASTER FLOW – PANEL 8/12: Slope-measure portfolio

WATER -> surface / subsurface drainage

GEOMETRY -> controlled excavation and toe protection

MATERIAL -> retaining protection bioengineering

LIMIT -> site-specific design maintenance residual risk

ASCII MASTER FLOW – PANEL 9/12: Route and settlement resilience

ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL

ALTERNATIVE ACCESS + SAFE STOPPAGE / CLOSURE

ASSEMBLY AREA + EVACUATION ROUTE

HOSPITAL POWER TELECOM WATER CONTINUITY

ASCII MASTER FLOW – PANEL 10/12: Warning-to-action chain

MONITOR -> ASSESS -> FORECAST / BULLETIN

AUTHORISE -> COMMUNICATE UNCERTAINTY

CLOSE ROUTE / EVACUATE -> SAFE DESTINATION

VERIFY RECEIPT ACTION AND FEEDBACK

MUST REMEMBER: Landslide, avalanche and GLOF risk in the Himalaya joins slope, geology,...

ASCII MASTER FLOW – PANEL 11/12: Development-risk balance

CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS

VERSUS

SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK

VERDICT -> RISK-INFORMED SITING NOT BLANKET CLAIMS

CLOSE DISTINCTION: Susceptibility is not probability; trigger is not root vulnerability;...

ASCII MASTER FLOW – PANEL 12/12: Mountain-risk answer spine

CLASSIFY PROCESS -> SEPARATE CONDITION FROM TRIGGER

DISTINGUISH SUSCEPTIBILITY HAZARD AND RISK

ASSIGN GSI DGRE NRSC CWC AND LOCAL ACTION

COMBINE ZONING DRAINAGE ROUTES WARNING EVACUATION + LIMITS

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Source event date, location, rainfall,...